

Problem 3

Problem 1 Circle the correct answer in each part:

- (a) Consider the limit $\lim_{x \rightarrow 0} (\cos(x))^{\sin(x)}$.
- (i) Evaluate the limit.
 - i. the limit DNE
 - ii. e
 - iii. 1
 - iv. ∞
 - v. $-\infty$
 - vi. 0
 - vii. none of the previous answers is correct
 - (ii) What Limit Law, rule or technique did you use to find this limit?
 - i. The Squeeze Theorem;
 - ii. L'Hôpital's Rule;
 - iii. The Product Law;
 - iv. evaluated the function at $x = 0$, since the function is continuous at $x = 0$;
 - v. none of the previous answers is correct
- (b) Evaluate the limit $\lim_{x \rightarrow 4^-} \frac{\ln(x)}{x - 4}$.
- (i) the limit DNE
 - (ii) e
 - (iii) 1
 - (iv) ∞
 - (v) $-\infty$
 - (vi) 0
 - (vii) none of the previous answers is correct
- (c) Evaluate the limit $\lim_{x \rightarrow \infty} \frac{\ln(x)}{x - 4}$.
- (i) the limit DNE
 - (ii) e

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- (iii) 1
- (iv) ∞
- (v) $-\infty$
- (vi) 0
- (vii) *none of the previous answers is correct*

(d) Consider the limit

$$\lim_{h \rightarrow 0} \frac{(2+h)^3 - 8}{h} = f'(2).$$

Determine the function f .

- (i) *such a function DNE;*
- (ii) $f(x) = x^3$;
- (iii) $f(x) = (2+x)^3$;
- (iv) $f(x) = \frac{(2+x)^3}{x}$;
- (v) *none of the previous answers is correct*

(e) Consider the limit: $\lim_{x \rightarrow 0^+} \left(\frac{\sin(x)}{x} \right)^{|\ln(x)|}$.

Determine the form of this limit.

- (i) $\frac{0}{0}$;
- (ii) $\frac{\infty}{\infty}$;
- (iii) 1^0 ;
- (iv) 0^0 ;
- (v) 1^∞ ;
- (vi) ∞^∞ ;
- (vii) *none of the previous answers is correct*